Monty Hall: Probability After a Reveal

Week 10 — a sample space, a host rule, and simulations

COMSC 2043 • Discrete Structures and Critical Thinking

October 21, 2026

State the rules before calculating

- Three doors; one car, two goats.
- You choose door 1.
- The host knows the car location, opens an unchosen goat door, and chooses uniformly if both doors are goats.
- You may stay with door 1 or switch to the remaining closed door.

The nine equally likely cases

Each ordered pair (your pick, car location) has probability 1/9.

	Car 1	Car 2	Car 3
Pick 1	H opens 2 or 3	H opens 3	H opens 2
Pick 2	H opens 3	H opens 1 or 3	H opens 1
Pick 3	H opens 2	H opens 1	H opens 1 or 2

By symmetry, the three rows have identical stay/switch outcomes.

The three symmetric cases

Assume your pick is door 1.

Car location	Host opens	Switch outcome
1	2 or 3 (equally)	Lose
2	3	Win
3	2	Win

Stay wins in one of the three cases; switching wins in two.

Conditional probability is a ratio inside an event

For events A, B with $P(B) > 0$, definition: $P(A \mid B) = P(A \cap B)/P(B)$.

After a host reveal, the relevant sample space is the set of cases compatible with that reveal and the host's rule.

The denominator changes when we condition.

Bayes: host opens door 3

Let C_i mean car behind door i; let O_3 mean host opens door 3.

$P(C_1) = P(C_2) = P(C_3) = 1/3$.
$P(O_3 \mid C_1) = 1/2$, $P(O_3 \mid C_2) = 1$, $P(O_3 \mid C_3) = 0$.

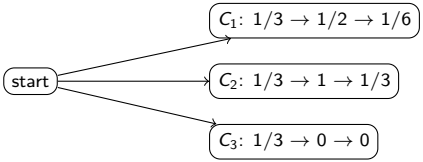
$P(O_3) = \sum_i P(O_3 \mid C_i)P(C_i) = 1/6 + 1/3 = 1/2$.

$P(C_1 \mid O_3) = ((1/2)(1/3))/(1/2) = 1/3$; hence $P(C_2 \mid O_3) = 2/3$. Switch.

Bayes denominator: add the paths that show door 3

Let O_3 mean the host opens door 3. The probability paths are: C_1 contributes $(1/3)(1/2) = 1/6$; C_2 contributes $(1/3)(1) = 1/3$; C_3 contributes $(1/3)(0) = 0$. Thus

$$P(O_3) = 1/6 + 1/3 + 0 = 1/2.$$



n doors: one closed alternative

You pick one of n doors. The first pick is correct with probability $1/n$.
A knowing host opens $n-2$ goats, leaving one alternative.
Switch wins exactly when the first pick was wrong: $P(\text{switch wins}) = (n-1)/n$.
For $n = 100$, this is $99/100$.

Open k goats; switch to one chosen survivor

Claim: With a uniform choice among survivors, each gets an equal share.

$$P(\text{first pick wrong}) = (n-1)/n \qquad \text{there are } n-1 \text{ wrong doors}$$

$$\text{alternatives left} = n-1-k \qquad \text{the host opens } k \text{ goats}$$

$$P(\text{win on specified door}) = \frac{(n-1)/n}{n-1-k} \qquad \text{split wrong-pick chance evenly}$$

$$= \frac{n-1}{n(n-1-k)} \qquad \text{simplify}$$

$$(n,k) = (5,2) \Rightarrow P = 4/10 \qquad \text{each of two doors gets } 0.4$$

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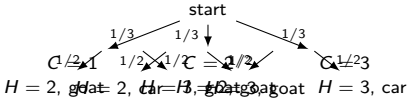
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Monty Fall: a random host

Claim: Given a goat is shown, staying and switching are equally likely to win.

$P(C=1) = P(C=2) = P(C=3) = 1/3$	car is equally likely
$P(G \mid C=1) = 1, \quad P(G \mid C=2) = P(G \mid C=3) = 1/2$	random host avoids car
$P(G) = 1/3 + (1/3)(1/2) + (1/3)(1/2) = 2/3$	add goat-showing branches
$P(C=1 \mid G) = (1/3)/(2/3) = 1/2$	stay wins only when car is picked
$P(C \neq 1 \mid G) = ((1/3)(1/2) + (1/3)(1/2))/(2/3) = 1/2$	switch wins on other branches



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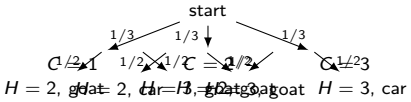
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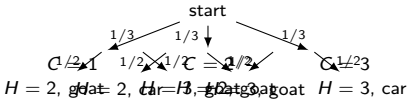
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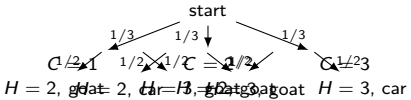


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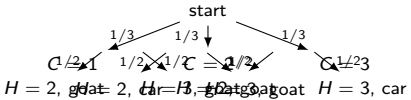
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Monty Crawl: a leftmost-door rule

Claim: Switching wins with probability $2/3$ before the reveal.

$P(C=1)=P(C=2)=P(C=3)=1/3$	three equally likely locations
$C=1 \Rightarrow H=2$, switch loses	first pick has the car
$C=2 \Rightarrow H=3$, switch wins	host opens leftmost eligible door
$C=3 \Rightarrow H=2$, switch wins	host opens the remaining goat
$P(\text{switch wins})=0+1/3+1/3=2/3$	add winning cases

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$P(C=1) = P(C=2) = P(C=3) = 1/3$	three equally likely locations
$C = 2 \Rightarrow H = 3$, switch wins	host opens leftmost eligible door
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Three Prisoners

Claim: Hearing that B is not pardoned leaves A at 1/3 and C at 2/3.

pardoned	P(case)	P(guard says B)	P(joint)
A	1/3	1/2	1/6
B	1/3	0	0
C	1/3	1	1/3

$P(\text{guard says B}) = 1/6 + 0 + 1/3 = 1/2$ add the B-name paths

$P(A \mid \text{guard says B}) = (1/6)/(1/2) = 1/3$ A's chance is unchanged

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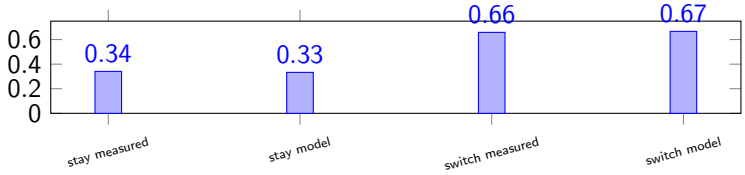
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Simulation checks a model's long-run rate

The simulator repeats the stated random experiment. For N independent trials, the law of large numbers says the observed fraction tends toward the model probability as N grows.

It does not prove the assumptions or promise a close result for a small N. Seeded measurement (10,000 games): stay $3412/10000 = 0.3412$; switch $6588/10000 = 0.6588$.



Playground: Buffon's hot dogs

Claim: The crossing rate estimates π when segment length equals line spacing.

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\begin{aligned}
P(\text{cross} \mid \theta) &= (\ell/t)\sin\theta && \text{center must be near a line} \\
p &= \frac{1}{\pi}\int_0^{\pi} (\ell/t)\sin\theta \, d\theta && \text{average over orientations} \\
&= 2\ell/(\pi t) && \text{evaluate the integral} \\
\widehat{p} &= \text{crossings}/N && \text{measure the fraction} \\
\widehat{\pi} &= 2\ell/(t\widehat{p}) && \text{solve for } \pi
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Playground: darts and birthdays

Claim: Area and complementary counting give exact model probabilities.

$$P(\text{dart hits}) = \pi/4 \qquad \text{quarter-circle area over square area}$$

$$\widehat{\pi} = 4(\text{hits}/N) \qquad \text{rearrange the measured rate}$$

$$P(\text{no birthday match}) = \prod_{i=0}^{n-1}(365 - i)/365 \qquad \text{multiply no-match chances}$$

$$P(\text{a match}) = 1 - \prod_{i=0}^{n-1}(365 - i)/365 \qquad \text{take the complement}$$

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Playground: collecting every coupon

Claim: Collecting n equally likely types takes nH_n draws on average.

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\begin{aligned}
P(\text{new type} \mid k \text{ held}) &= (n-k)/n && n-k \text{ unseen types} \\
E(W_k) &= n/(n-k) && \text{geometric wait is reciprocal} \\
E(T) &= \sum_{k=0}^{n-1} n/(n-k) && \text{add stage waits} \\
&= n \sum_{j=1}^{n} 1/j = nH_n && \text{rewrite as harmonic sum} \\
E(T)_{n=10} &\approx 29.29 && \text{compare with simulated mean}
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Secretary rule: derive the finite-n chance

Set $r = \lfloor n/e \rfloor$. Skip the first r applicants and remember their best. If the overall best is at position $j > r$, the rule selects it exactly when the best of the first $j-1$ positions lies among the first r. That probability is $r/(j-1)$, while the overall best is at j with probability $1/n$. Therefore

$$P(\text{success}) = \sum_{j=r+1}^{n} \frac{r}{n(j-1)} \longrightarrow 1/e.$$

For $n = 100$, $r = 36$ and the finite sum is approximately 0.3710 (the limit is $1/e \approx 0.3679$).

What to remember

- A probability answer depends on the sample space and the host's rule.
- A reveal can change conditional probabilities; it does not erase the original choice.
- Derive the model first, then use simulation to check its numerical behavior.